

# Accounting Ratios

Class 12 Accountancy — Financial Statement Analysis (Revision Notes)

|                             |                                                                          |
|-----------------------------|--------------------------------------------------------------------------|
| <b>Subject:</b>             | Class 12 Accountancy (Part B)                                            |
| <b>Modes of Expression:</b> | Pure Ratio (e.g., 2:1), Percentage (%), Times (e.g., 4 times), Fraction. |
| <b>Classification:</b>      | 1. Liquidity   2. Solvency   3. Activity / Turnover   4. Profitability   |

## 1. Liquidity Ratios (Short-Term Solvency)

Measures the ability of the enterprise to meet its short-term financial obligations.

| Ratio Name              | Formula                                     | Ideal / Key Notes                                                                                                                          |
|-------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Current Ratio        | <b>Current Assets ÷ Current Liabilities</b> | <b>Ideal: 2 : 1</b><br><i>CA = Inventory + Trade Receivables + Cash + Short-term loans/advances. (Exclude Loose tools from Inventory).</i> |
| 2. Quick / Liquid Ratio | <b>Quick Assets ÷ Current Liabilities</b>   | <b>Ideal: 1 : 1</b><br><i>Quick Assets = Current Assets – (Inventory + Prepaid Expenses).</i>                                              |

## 2. Solvency Ratios (Long-Term Solvency)

Measures the ability of the firm to meet its long-term liabilities and interest commitments.

| Ratio Name                    | Formula                                                                     | Key Components                                                                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Debt to Equity Ratio       | <b>Debt ÷ Shareholders' Funds</b>                                           | <b>Ideal: 2 : 1</b> <ul style="list-style-type: none"><li>Debt = Non-Current Liabilities (Debentures + Long-Term Loans).</li><li>Equity = Share Capital + Reserve &amp; Surplus.</li></ul> |
| 2. Total Assets to Debt Ratio | <b>Total Assets ÷ Long-Term Debt</b>                                        | Higher ratio indicates higher safety for long-term lenders.                                                                                                                                |
| 3. Proprietary Ratio          | <b>Proprietors' Funds ÷ Total Assets</b>                                    | Expressed as pure ratio or percentage. Measures owner's contribution to total assets.                                                                                                      |
| 4. Interest Coverage Ratio    | <b>Profit before Interest &amp; Tax (PBIT) ÷ Interest on Long-Term Debt</b> | Expressed in <b>Times</b> . Higher ratio means safer interest payment ability.                                                                                                             |

### 3. Activity / Turnover Ratios

Measures how efficiently the resources/assets are being utilized by the firm. (Expressed in **Times**).

| Ratio Name                          | Formula                                                                       | Key Components                                                                                                                                                                                    |
|-------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Inventory Turnover Ratio         | $\text{Cost of Revenue from Operations (COGS)} \div \text{Average Inventory}$ | <ul style="list-style-type: none"><li>• COGS = Revenue from Ops – Gross Profit.</li><li>• Avg Inventory = <math>(\text{Opening} + \text{Closing Inv}) \div 2</math>.</li></ul>                    |
| 2. Trade Receivables Turnover Ratio | $\text{Credit Revenue from Operations} \div \text{Average Trade Receivables}$ | <ul style="list-style-type: none"><li>• Avg TR = <math>(\text{Opening TR} + \text{Closing TR}) \div 2</math>.</li><li>• Debt Collection Period = <math>365 / \text{TR Turnover}</math>.</li></ul> |
| 3. Trade Payables Turnover Ratio    | $\text{Net Credit Purchases} \div \text{Average Trade Payables}$              | <ul style="list-style-type: none"><li>• Avg TP = <math>(\text{Opening TP} + \text{Closing TP}) \div 2</math>.</li><li>• Debt Payment Period = <math>365 / \text{TP Turnover}</math>.</li></ul>    |
| 4. Working Capital Turnover Ratio   | $\text{Revenue from Operations} \div \text{Working Capital}$                  | <ul style="list-style-type: none"><li>• Working Capital = Current Assets – Current Liabilities.</li></ul>                                                                                         |

### 4. Profitability Ratios

Measures the overall operational efficiency and earning capacity of the business. (Expressed in **Percentage %**).

| Ratio Name                    | Formula                                                                               | Key Components                                                                                                                                                    |
|-------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Gross Profit Ratio         | $(\text{Gross Profit} \div \text{Revenue from Operations}) \times 100$                | Gross Profit = Net Sales – COGS.                                                                                                                                  |
| 2. Operating Ratio            | $[(\text{COGS} + \text{Operating Expenses}) \div \text{Revenue from Ops}] \times 100$ | Operating Expenses = Employee benefit expenses + Selling & admin costs.                                                                                           |
| 3. Operating Profit Ratio     | $(\text{Operating Profit} \div \text{Revenue from Operations}) \times 100$            | <ul style="list-style-type: none"><li>• Operating Profit = Net Sales – Operating Cost.</li><li>• Note: Operating Ratio + Operating Profit Ratio = 100%.</li></ul> |
| 4. Net Profit Ratio           | $(\text{Net Profit After Tax} \div \text{Revenue from Operations}) \times 100$        | Measures overall profitability after all expenses.                                                                                                                |
| 5. Return on Investment (ROI) | $(\text{Profit Before Interest \& Tax} \div \text{Capital Employed}) \times 100$      | Capital Employed = Shareholder Funds + Long Term Debt (or Non-Current Assets + Working Capital).                                                                  |